

Dilated Cardiomyopathy 2E (CMD2E): A Comprehensive Disease Characterization

Disease: Dilated Cardiomyopathy 2E (CMD2E) **MONDO ID:** MONDO:0030366 · **OMIM** **Phenotype:** #619492 · **Gene:** JPH2 (junctophilin-2) **Category:** Mendelian (autosomal recessive)

Summary

Dilated Cardiomyopathy 2E (**CMD2E**; OMIM #619492, MONDO:0030366) is a rare, autosomal **recessive**, non-syndromic form of dilated cardiomyopathy caused by **biallelic loss-of-function (LOF) mutations in JPH2**, the gene encoding **junctophilin-2** (chromosome 20q13.12; OMIM 605267; HGNC:14179; UniProt Q9BR39). *The disease was established by two independent recessive kindreds: a Finnish family with a homozygous nonsense variant p.Q428X (c.1282C>T) in which the proband was diagnosed at age 3 and transplanted at age 4 (PMID: 30384889), and consanguineous Iranian families sharing a founder haplotype carrying a homozygous 1-bp insertion p.E641* (c.1920dupT)* with neonatal DCM and childhood death (PMID: 31227780).*

The mechanistic core of the disease is the **cardiac dyad**. Junctophilin-2 physically tethers the T-tubule sarcolemma to the junctional sarcoplasmic reticulum (jSR), maintaining the ~12–15 nm dyadic cleft that positions L-type Ca²⁺ channels (CACNA1C/LTCC) opposite ryanodine receptors (RyR2) so that calcium-induced calcium release (CICR) can drive contraction. When both *JPH2* alleles are lost, the dyad cannot assemble, excitation–contraction (EC) coupling fails, and the ventricle dilates with progressively impaired systolic function. This is corroborated by mouse genetics: germline *Jph2*-null embryos die immediately after the heart begins to beat, with deficient junctional membrane complexes and asynchronous Ca²⁺ transients (PMID: 10949023), while cardiac-specific knockdown produces contractile failure and heart failure (PMID: 21339484).

Clinically, CMD2E presents in the **neonatal period or early childhood** with rapidly progressive systolic heart failure, frequently requiring mechanical circulatory support or **heart transplantation**. There is **no CMD2E-specific approved therapy**; management follows standard guideline-directed pediatric heart-failure care, with **AAV9-mediated JPH2 gene**

replacement as a strong preclinical rationale (PMID: 27760414). CMD2E is mechanistically and genetically distinct from the *dominant* JPH2 **missense** hypertrophic cardiomyopathy (CMH17), which acts through altered RyR2 regulation rather than protein loss.

1. Disease Information

Overview. CMD2E is a Mendelian, gene-defined subtype of dilated cardiomyopathy (DCM). DCM is characterized by left ventricular (or biventricular) dilation and systolic dysfunction not explained by abnormal loading conditions or coronary disease. CMD2E is the specific entity attributable to biallelic *JPH2* loss-of-function.

Key identifiers.

Resource	Identifier
OMIM (phenotype)	#619492 (CARDIOMYOPATHY, DILATED, 2E; CMD2E)
OMIM (gene)	*605267 (JPH2)
MONDO	MONDO:0030366
HGNC	HGNC:14179 (JPH2)
NCBI Gene	57158
Ensembl	ENSG00000149596
UniProt	Q9BR39
Cytoband	20q13.12 (GRCh38 chr20:44,106,590–44,187,188)
ICD-10 (parent DCM)	I42.0
ICD-11 (parent DCM)	BC43.0
MeSH (parent DCM)	D002311
Orphanet	Under familial isolated DCM (ORPHA:154)

Synonyms / alternative names. "Dilated cardiomyopathy 2E"; "CMD2E"; "JPH2-related dilated cardiomyopathy"; "junctophilin-2-related recessive dilated cardiomyopathy." Note that "CMD2E" (with numeric-letter suffix) denotes the OMIM molecular-genetic subtype and should not be confused with dominant JPH2 disease.

Information source type. Evidence is derived from **aggregated disease-level resources** (OMIM, MONDO, gnomAD) combined with **individual-patient case reports and small kindreds** (two founding families plus subsequent case reports), and from **model-organism and in vitro** mechanistic studies. There is no large EHR-derived cohort specific to CMD2E owing to its rarity.

2. Etiology

Primary cause — genetic. CMD2E is caused by **homozygous or compound heterozygous loss-of-function variants in *JPH2***. Two independent recessive kindreds established causality (Finding F001):

- **p.Q428X (c.1282C>T)**, homozygous nonsense, Finnish family — proband diagnosed at age 3, transplanted at age 4 ([PMID: 30384889](#)). The paper identifies JPH2 among calcium-signaling genes in a severe childhood cardiomyopathy cohort: "*calcium signaling (JPH2, CALM1, CACNA1C).*"
- **p.E641* (c.1920dupT)**, homozygous 1-bp insertion creating a premature stop, consanguineous Iranian families on a shared founder haplotype ([PMID: 31227780](#)): "*identified an Iranian patient with dilated cardiomyopathy (DCM) as a carrier of a novel, homozygous single nucleotide insertion in JPH2 resulting in a stop codon (JPH2-p.E641*).*"

Genetic risk factors. The obligate risk factor is inheritance of **two** LOF *JPH2* alleles. **Consanguinity** and **founder effects** substantially elevate risk (the Iranian p.E641* variant on a shared haplotype). Heterozygous carriers are asymptomatic (see §9), consistent with a recessive mechanism.

Environmental risk factors. No specific environmental trigger is established as causal for CMD2E; the disease is monogenic. General DCM environmental modifiers (viral myocarditis, toxins, alcohol) are not implicated in the biallelic-JPH2 entity, though they could theoretically aggravate any myocardial reserve deficit.

Protective factors. No genetic or environmental protective factors are specifically documented for CMD2E. By inference, retention of even one functional *JPH2* allele is "protective" (carriers are unaffected), reflecting haplosufficiency.

Gene–environment interactions. None specifically documented. The dominant driver is genotype (biallelic LOF); environmental contribution appears minimal relative to the primary lesion.

3. Phenotypes

CMD2E presents with the core phenotype of **dilated cardiomyopathy**: left ventricular dilation and impaired systolic function (reduced ejection fraction) producing **congestive heart failure**, with **arrhythmia risk** and risk of **premature death** (Finding F006). Progression is **rapid**, frequently to transplant-dependent end-stage failure.

Phenotype	Type	HPO term (suggested)	Onset	Severity / progression	Frequency
Dilated cardiomyopathy	Clinical sign (imaging)	HP:0001644	Neonatal–early childhood	Severe, progressive	Defining (all patients)
Left ventricular systolic dysfunction / reduced EF	Laboratory/imaging abnormality	HP:0005162 / HP:0012664	Neonatal–early childhood	Severe, progressive	Very frequent
Congestive heart failure	Clinical sign	HP:0001635	Neonatal–early childhood	Severe	Very frequent
Ventricular arrhythmia / arrhythmia	Clinical sign	HP:0004308 / HP:0011675	Childhood	Variable	Reported risk
Sudden cardiac death / premature death	Outcome	HP:0001645 / HP:0001663	Childhood	—	Reported

Phenotype characteristics. *Age of onset:* neonatal to early childhood (congenital/pediatric). *Severity:* severe. *Progression:* rapid and progressive. *Frequency:* the DCM/systolic-failure phenotype is fully penetrant in reported biallelic patients. Supporting quote ([PMID: 31227780](#)): "A second Iranian family with consanguineous parents hosting an identical heterozygous variant had 2 children die in childhood from cardiac failure."

Quality-of-life impact. Not formally measured with standardized instruments (EQ-5D/SF-36/PROMIS) in this rare disease, but the burden is profound: infants/children experience heart-failure symptoms (feeding difficulty, failure to thrive, dyspnea, exercise intolerance), hospitalization, need for mechanical support, and transplantation with lifelong immunosuppression.

4. Genetic / Molecular Information

Causal gene. *JPH2* — junctophilin-2 (OMIM *605267; HGNC:14179; NCBI Gene 57158; Ensembl ENSG00000149596; UniProt Q9BR39; 20q13.12). Finding F007 verified these identifiers via mygene.info.

Pathogenic variants (CMD2E, recessive).

Variant (protein)	cDNA	Type	Zygosity	Population	Reference
p.Q428X	c.1282C>T	Nonsense (LOF)	Homozygous	Finnish	PMID: 30384889
p.E641*	c.1920dupT	Frameshift/ stop-gain (LOF)	Homozygous (founder)	Iranian / Greater Middle East	PMID: 31227780

Variant classification. Both founding variants are truncating **loss-of-function** alleles (nonsense / frameshift-stopgain) and are consistent with **Pathogenic** classification under ACMG/AMP (null variant in a gene where LOF is a known disease mechanism, segregation, rarity/absence in controls). Additional recessive frameshift cases have subsequently been reported in pediatric DCM (e.g., [PMID: 41919412](#)).

Variant type/class. Truncating (nonsense, frameshift → premature termination codon); predicted to trigger nonsense-mediated decay or produce truncated non-functional protein → **loss of function**.

Allele frequency (population databases). gnomAD constraint for *JPH2* (Finding F004): **pLI $\approx 3.9 \times 10^{-9}$, LOEUF (oe_lof_upper) = 0.89**, observed/expected LOF = **38/55.9 (oe_lof 0.68)** — i.e., heterozygous LOF is **tolerated**, consistent with unaffected carriers and a recessive disease requiring biallelic loss. LOF variants are rare overall (0.04%) but enriched in Greater Middle Eastern (GME) individuals (0.21%) (**PMID: 31227780**): *"Worldwide, 1.45% of healthy individuals hosted a rare JPH2 variant with a significantly higher proportion among GME individuals (4.45%); LOF variants were rare overall (0.04%) yet were most prevalent in GME (0.21%)."*

Somatic vs germline. Germline (inherited). No somatic mechanism is relevant.

Functional consequences. Loss of function (haploinsufficiency is tolerated; biallelic loss is pathogenic). This contrasts sharply with the *dominant missense* *JPH2* variants causing hypertrophic cardiomyopathy/arrhythmia (e.g., A405S — **PMID: 28393127**; E169K causing atrial fibrillation via impaired RyR2 stabilization — **PMID: 23973696**), which act by altered/reduced RyR2 regulation rather than complete protein loss.

Modifier genes. None specifically established for CMD2E. Other calcium-handling/dyadic genes (RYR2, CACNA1C, CALM1, BIN1, CMYA5) are biologically plausible modifiers but unproven in this disorder.

Epigenetic information / chromosomal abnormalities. No disease-specific DNA-methylation, histone-modification, or large-scale chromosomal abnormality (aneuploidy/translocation) is documented for CMD2E. The disease is a small-scale sequence (point/indel) disorder.

5. Environmental Information

CMD2E is a **monogenic recessive** disorder; **no environmental, lifestyle, or infectious agents are established as causal**. Unlike acquired DCM (viral myocarditis, alcohol, chemotherapy toxins), the biallelic-*JPH2* entity arises from the genetic lesion. Consanguinity is the principal non-molecular contributor to disease occurrence at the population level, by increasing the probability of biallelic inheritance of rare founder LOF alleles (§9). Environmental cardiac stressors are, at most, plausible aggravators of already-compromised myocardial reserve, not initiators.

6. Mechanism / Pathophysiology

Ordered causal chain

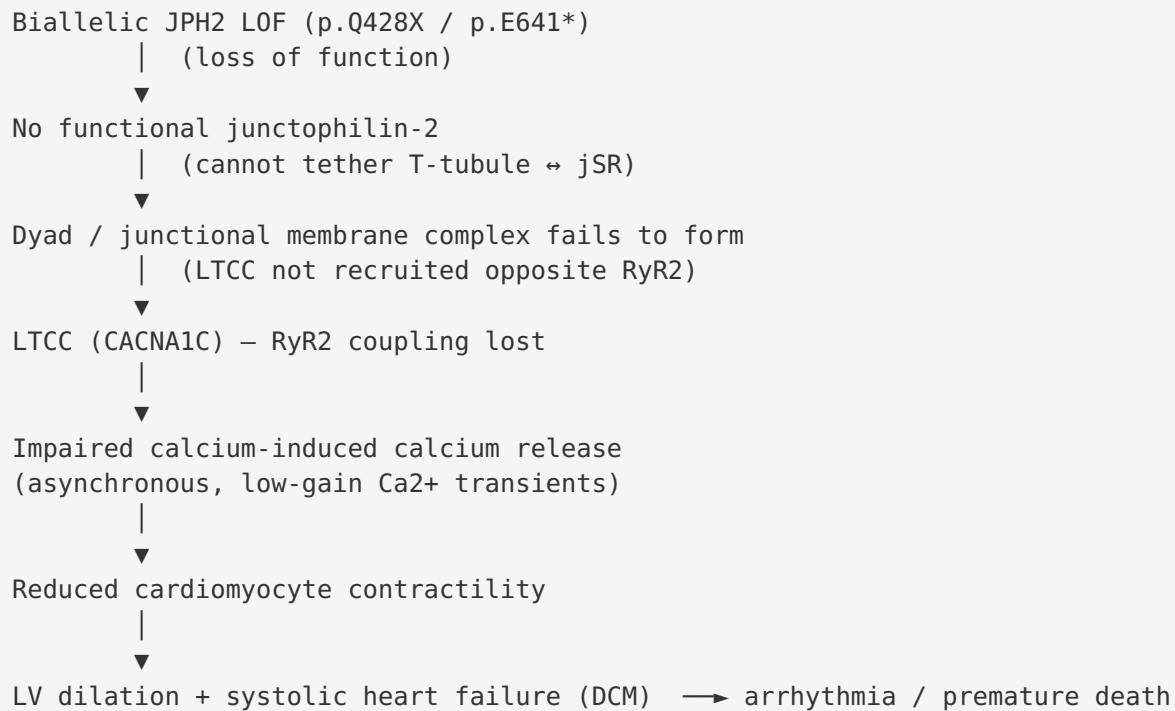
1. **Biallelic *JPH2* loss-of-function mutation** (e.g., p.Q428X, p.E641*) **leads to** absence/severe reduction of functional junctophilin-2 protein in cardiomyocytes (loss of function; both alleles null).
2. Loss of junctophilin-2 **results in** failure to tether the T-tubule sarcolemma to the junctional sarcoplasmic reticulum, so the **cardiac dyad / junctional membrane complex cannot assemble or be maintained** (the ~12–15 nm dyadic cleft is lost or disordered).
Demonstrated in mouse *Jph2*-null myocytes ([PMID: 10949023](#)).
3. Without an intact dyad, **L-type Ca^{2+} channels (CACNA1C/LTCC) are no longer recruited to and juxtaposed with RyR2** — the JPH2 "joining region" that directly binds the LTCC $\alpha 1\text{C}$ subunit is lost ([PMID: 33092464](#)), which **leads to** loss of the tight spatial coupling required for CICR.
4. Disrupted LTCC–RyR2 coupling **results in** impaired/asynchronous **calcium-induced calcium release** and reduced EC-coupling gain — abnormal, spatially and temporally desynchronized Ca^{2+} transients (*demonstrated* in null and knockdown myocytes; [PMID: 10949023](#), [PMID: 21339484](#)).
5. Deficient Ca^{2+} transients **lead to** reduced cardiomyocyte contractility (weak systole).
6. Chronic contractile insufficiency **results in ventricular dilation and systolic heart failure** (the DCM phenotype) — *inferred* progression from cellular to organ level, supported by cardiac-knockdown mice developing heart failure ([PMID: 21339484](#)).
7. Ventricular remodeling and abnormal Ca^{2+} handling **lead to arrhythmia risk and premature death** (*inferred* from clinical course and RyR2 dysregulation literature).

Branch (stress-adaptive signaling). In parallel with structural loss, proteolytic cleavage of JPH2 (by calpain, [PMID: 30409805](#); by MMP-2, [PMID: 31506724](#)) generates an N-terminal fragment (JP2NT) that translocates to the nucleus as a **stress-adaptive transcriptional regulator**. This branch is most relevant to acquired heart failure, but it underscores that JPH2 has both a **structural** (dyad-tethering) and a **signaling** role; in CMD2E, complete germline loss abolishes both. Supporting quote ([PMID: 30409805](#)): "*After cardiac stress, JP2 is cleaved by the calcium ion-dependent protease calpain, which disrupts the E-C coupling ultrastructural machinery and drives heart failure progression.*"

Category detail

- **Molecular pathways.** Cardiac **excitation–contraction coupling** and **calcium-induced calcium release** (CICR) at the dyad; LTCC (CACNA1C) → RyR2 signaling axis. GO biological processes: **GO:0006942** (regulation of striated muscle contraction), **GO:0055117** (regulation of cardiac muscle contraction), **GO:0010881** (regulation of cardiac muscle contraction by regulation of the release of sequestered calcium ion), **GO:0051209** (release of sequestered calcium ion into cytosol). Supporting quote ([PMID: 33092464](#)): *"The interaction between LTCC and the joining region in JPH2 facilitates dyad assembly and maintains normal CICR in cardiomyocytes."*
- **Cellular processes.** Impaired myocyte contraction; disrupted junctional membrane complex biogenesis; abnormal intracellular Ca²⁺ homeostasis. In the acquired/proteolysis branch: stress-adaptive transcriptional reprogramming.
- **Protein dysfunction.** Loss of function — truncated/absent junctophilin-2; loss of the membrane-tethering MORN motifs and the LTCC-binding "joining region."
- **Metabolic changes.** No primary metabolic defect; secondary bioenergetic strain accompanies heart failure (not specific to CMD2E).
- **Immune involvement.** Not a primary feature; CMD2E is not autoimmune/inflammatory in origin.
- **Tissue-damage mechanisms.** Progressive myocardial dysfunction and dilation; interstitial/ collagen remodeling accompanies advanced failure (t-tubule/dyad disorganization associated with collagen deposition in failing human myocardium — [PMID: 34690801](#)).
- **Biochemical abnormality.** Defective calcium-channel coupling (LTCC–RyR2 microdomain failure) — an ion-handling/microdomain defect rather than an enzyme deficiency.
- **Cell types & GO cellular components.** Cell type: **ventricular cardiac muscle cell (CL:0002131)** / cardiac muscle cell (CL:0000746). Cellular components: **cardiac dyad, T-tubule (GO:0033268 / GO:0014801)**, **junctional sarcoplasmic reticulum membrane (GO:0014701)** / sarcoplasmic reticulum (GO:0016529), **Z-disc (GO:0030018)**.

Mechanistic schematic



7. Anatomical Structures Affected

- **Organ level (primary).** Heart (UBERON:0000948), specifically the **ventricular myocardium / cardiac ventricle** (UBERON:0002349 / UBERON:0002082); left ventricle predominant with possible biventricular involvement. **Body system:** cardiovascular.
- **Secondary organ involvement.** Downstream congestive effects of heart failure (pulmonary congestion, hepatic congestion, systemic hypoperfusion) — complications rather than primary sites.
- **Tissue/cell level.** Cardiac (striated) muscle tissue; ventricular cardiac muscle cells / cardiomyocytes (CL:0002131 / CL:0000746).
- **Subcellular level.** The cardiac dyad — the junction of T-tubule (GO:0033268) and junctional sarcoplasmic reticulum (GO:0014701 / GO:0016529); the Z-disc (GO:0030018); sarcolemma–SR contact microdomain.
- **Localization / lateralization.** Bilateral/global ventricular process (not lateralized); a diffuse cardiomyocyte-autonomous defect rather than a focal lesion.

8. Temporal Development

- **Onset. Neonatal to early-childhood** (congenital/pediatric). Onset pattern is early and can be insidious to subacute, progressing to overt decompensated heart failure. In the founding kindreds, the Finnish proband was diagnosed at age 3 ([PMID: 30384889](#)) and Iranian children died in childhood from cardiac failure ([PMID: 31227780](#)).
- **Progression. Rapid and progressive**, advancing to **end-stage** systolic failure. Disease course is progressive (not relapsing-remitting), typically culminating in transplant dependence within a few years of diagnosis (Finnish proband transplanted at age 4).
- **Duration.** Chronic and lifelong from a genetic standpoint; without transplantation the natural course is early fatal.
- **Patterns / remission.** No spontaneous remission is expected; "remission" is achieved only via transplantation (replacement of the affected organ). Critical intervention window is early — before irreversible ventricular remodeling and end-organ compromise.

9. Inheritance and Population

Inheritance pattern. Autosomal recessive (biallelic LOF required). Established by homozygous variants segregating with disease in two independent consanguineous/founder kindreds (Finding F001).

Penetrance / expressivity. In reported **biallelic** individuals, penetrance of the DCM/heart-failure phenotype appears **complete/high**; **heterozygous carriers are unaffected**. Expressivity within biallelic patients is severe with early onset; the small number of families limits precise estimates.

Carrier frequency & founder effects. gnomAD shows *JPH2* tolerates heterozygous LOF (LOEUF 0.89; oe_lof 0.68), consistent with silent carriers (Finding F004). Worldwide ~1.45% of healthy individuals carry a rare *JPH2* variant, rising to 4.45% in GME populations; LOF alleles are rare (0.04%) but most prevalent in GME (0.21%) ([PMID: 31227780](#)). The Iranian **p.E641*** allele arose on a shared **founder haplotype**.

Consanguinity. A major contributor — recessive disease manifests when consanguineous unions bring together two copies of a rare founder LOF allele.

Anticipation / mosaicism. No evidence of genetic anticipation (not a repeat-expansion disorder). Germline mosaicism not specifically reported.

Epidemiology. CMD2E is **very rare** (an ultra-rare Mendelian subtype of DCM); precise prevalence/incidence figures are not established given the small number of reported families. It falls under the Orphanet umbrella of familial isolated DCM (ORPHA:154). Enrichment is expected in **consanguineous populations, particularly the Greater Middle East**.

Demographics. No strong sex bias is established (recessive, autosomal). Affected individuals are infants/children. Geographic clustering follows founder/consanguinity patterns (e.g., Iranian founder for p.E641*).

10. Diagnostics

Clinical/imaging. Diagnosis of the DCM phenotype rests on **echocardiography** (LV dilation, reduced ejection fraction) and **cardiac MRI** (chamber dimensions, systolic function, fibrosis on late gadolinium enhancement). **ECG** and Holter monitoring assess arrhythmia/conduction. **Biomarkers:** elevated **BNP/NT-proBNP** (heart-failure severity); **troponin** may be monitored. Endomyocardial biopsy is not required for diagnosis but would show non-specific myocyte changes/remodeling.

Genetic testing (definitive for CMD2E). Because CMD2E is defined molecularly, **genetic testing is the confirmatory diagnostic**: - **Cardiomyopathy/DCM multigene NGS panels** including *JPH2* are first-line. - **Whole-exome (WES) or whole-genome sequencing (WGS)** is highly useful, especially for early-onset/consanguineous pediatric DCM where recessive genes are implicated; the founding CMD2E variants were identified via broad sequencing in severe childhood cardiomyopathy cohorts ([PMID: 30384889](#)). - **Confirming biallelic status** (homozygous vs compound heterozygous, with parental segregation) is essential to distinguish recessive CMD2E from incidental heterozygous carriage. - Chromosomal microarray/karyotype/FISH/mtDNA/repeat-expansion testing are **not** indicated for this small-scale sequence disorder.

Clinical criteria & differential diagnosis. Standard DCM diagnostic criteria (dilation + systolic dysfunction not explained by loading/coronary disease). **Differential:** other genetic pediatric DCM (e.g., TTN, LMNA, FLNC, DES, DSP, TNNT2), syndromic/oligogenic DCM ([PMID:](#)

[42559185](#)), metabolic/mitochondrial cardiomyopathies, myocarditis, and — importantly — **dominant JPH2 hypertrophic cardiomyopathy (CMH17)**, which is phenotypically hypertrophic (not dilated) and mechanistically distinct.

Screening. For affected families: **cascade genetic testing** of relatives to identify carriers and at-risk future offspring; **carrier screening** in consanguineous couples with a founder allele; prenatal/preimplantation testing where a familial biallelic risk exists.

11. Outcome / Prognosis

Survival/mortality. Prognosis is **poor without transplantation**: both founding kindreds illustrate early end-stage disease — transplantation at age 4 in the Finnish proband ([PMID: 30384889](#)) and childhood death from cardiac failure in Iranian children ([PMID: 31227780](#)). With modern pediatric heart-failure care, outcomes for end-stage DCM are favorable: registry data show adolescents with DCM have excellent 3-year survival with either a HeartMate 3 VAD (94.4%) or transplant (95.6%) ([PMID: 42334151](#)): *"Adolescents with DCM treated with either HM3 or transplant had excellent 3-year outcomes."*

Morbidity/function. High morbidity from heart failure (hospitalizations, activity limitation, feeding/growth impairment in infants), device-related complications in those on mechanical support, and lifelong immunosuppression after transplant.

Complications. Progressive systolic failure, ventricular arrhythmia, thromboembolism, end-organ hypoperfusion, and sequelae of mechanical support/transplant.

Prognostic factors. Age/severity at presentation, degree of ventricular dysfunction, arrhythmia burden, and response to guideline-directed therapy. Recovery of native ventricular function is not expected given the fixed genetic dyadic defect; definitive treatment is organ replacement (or, prospectively, gene replacement).

12. Treatment

No CMD2E-specific approved therapy exists (Finding F005). Management is **standard pediatric dilated-cardiomyopathy / heart-failure care**:

- **Pharmacotherapy (guideline-directed medical therapy).** ACE inhibitors / ARBs, **beta-blockers, mineralocorticoid-receptor antagonists, diuretics**; contemporary HFrEF agents as age-appropriate. (Suggested NCIT concept classes: *Angiotensin-Converting Enzyme Inhibitor, Beta-Adrenergic Blocker, Diuretic, Aldosterone Antagonist*.)
- **Mechanical circulatory support. Ventricular assist devices** (e.g., HeartMate 3) as bridge-to-transplant or longer-term support; pediatric VAD outcomes are well documented ([PMID: 42622114](#), [PMID: 42334151](#)). (NCIT: *Ventricular Assist Device*.)
- **Orthotopic heart transplantation.** The definitive treatment for end-stage disease; the Finnish proband was transplanted at age 4. (NCIT: *Heart Transplantation*.)
- **Device therapy / arrhythmia management** as indicated (ICD for high arrhythmic risk).

Targeted / advanced therapeutics (investigational rationale). Because CMD2E is a **loss-of-function** disorder, **gene replacement** is the mechanistically ideal strategy. **AAV9-mediated JPH2 overexpression rescued cardiac contractility, preserved T-tubule structure, and normalized RyR2-mediated Ca²⁺ release** in a mouse (transverse aortic constriction) heart-failure model ([PMID: 27760414](#)): "*AAV9-mediated expression of JPH2 rescued cardiac contractility in mice subjected to TAC. AAV9-JPH2 also preserved T-tubule structure.*" This provides a strong preclinical rationale for **AAV9-JPH2 gene therapy** in CMD2E, though no human trial yet exists.

Pharmacogenomics. No gene-specific pharmacogenomic guidance is established for CMD2E.

Treatment outcomes. Supportive/GDMT slows progression but does not correct the primary defect; end-stage disease is managed with VAD/transplant, with excellent short-to-medium-term survival in modern pediatric practice ([PMID: 42334151](#)).

13. Prevention

- **Primary prevention.** No population-level primary prevention (monogenic disease). The principal lever is **reproductive genetic counseling** in families/populations carrying founder LOF alleles — particularly reducing recessive risk in consanguineous unions.

- **Secondary prevention. Cascade genetic screening** of relatives of probands to identify carriers and at-risk offspring; early echocardiographic surveillance in genetically at-risk children enables timely heart-failure therapy.
 - **Tertiary prevention.** Optimizing GDMT, arrhythmia risk management, and timely VAD/transplant to prevent complications and death in affected individuals.
 - **Genetic counseling & reproductive options.** For couples both carrying a *JPH2* LOF allele, recurrence risk is 25% per pregnancy; **preimplantation genetic testing** and **prenatal diagnosis** are options. Carrier screening is especially relevant in Greater Middle Eastern / consanguineous populations given founder enrichment ([PMID: 31227780](#)).
 - **Immunization / public health / prophylaxis.** Not applicable (non-infectious, monogenic).
-

14. Other Species / Natural Disease

- **Taxonomy / orthologs (Finding F007).** *JPH2* is conserved across vertebrates: **mouse *Jph2*** (NCBI Gene 59091; Ensembl ENSMUSG00000017817; UniProt Q9ET78), **zebrafish *jph2*** (NCBI Gene 553333; Ensembl ENSDARG00000028625). The essential role in cardiac dyad/EC-coupling formation is conserved.
 - **Natural disease in other species.** No well-characterized naturally occurring biallelic-JPH2 dilated cardiomyopathy is documented in companion animals or wildlife in the reviewed literature; disease knowledge derives from engineered models (§15).
 - **Comparative biology / evolutionary conservation.** The dyad is an evolutionarily conserved cardiomyocyte structure, and junctophilin's tethering function is conserved, making cross-species models highly informative. (For background on dyad evolution, see [PMID: 36189805](#).)
 - **Zoonotic potential.** Not applicable.
-

15. Model Organisms

CMD2E biology is strongly supported by mouse genetics (Finding F003):

Model	Manipulation	Key phenotype	Reference
Germline <i>Jph2</i> -null mouse	Global knockout	Embryonic lethal immediately after heartbeat onset; deficient junctional membrane complexes; abnormal, asynchronous Ca ²⁺ transients	PMID: 10949023 ; PMID: 12086916
Cardiac-specific shRNA JPH2 knockdown	Partial (postnatal) loss	Impaired contractility, fewer junctional membrane complexes, increased plasmalemma–SR distance variability, reduced EC-coupling gain, heart failure and increased mortality	PMID: 21339484
AAV9-JPH2 gene delivery (TAC HF model)	Overexpression/ rescue	Rescued contractility, preserved T-tubules, normalized RyR2-mediated Ca ²⁺ release	PMID: 27760414

Model types available. Mammalian (mouse) knockout, cardiac-specific knockdown, and AAV-based rescue; zebrafish orthologs enable developmental modeling. iPSC-derived cardiomyocytes from patients are a logical (in vitro) system to model biallelic human variants.

Phenotype recapitulation. Mouse models faithfully recapitulate the **core mechanism** — dyad loss, EC-coupling failure, abnormal Ca²⁺ handling, and contractile/heart failure — validating the human causal chain. Supporting quotes: "*JP-2 is abundantly expressed in the heart, and mutant mice lacking JP-2 exhibited embryonic lethality. Cardiac myocytes from the mutant mice showed deficiency of the junctional membrane complexes and abnormal Ca²⁺ transients*" (PMID: [10949023](#)); "*Cardiac-specific JPH2 knockdown resulted in impaired cardiac contractility, which caused heart failure and increased mortality*" (PMID: [21339484](#)).

Model limitations. Germline null is embryonic-lethal, so it cannot model the postnatal human course; knockdown produces partial rather than complete loss; TAC models pressure-overload HF rather than the primary congenital dyadic deficit. A conditional/humanized biallelic-LOF model or patient iPSC-cardiomyocytes would better capture the human CMD2E trajectory.

Mechanistic Model / Interpretation

CMD2E is best understood as a **structural dyadopathy of the cardiomyocyte**. Junctophilin-2 is the molecular "staple" that holds the T-tubule sarcolemma against the junctional SR; its "joining region" additionally recruits L-type Ca^{2+} channels into the dyad opposite RyR2. Removing both alleles removes the staple, the dyad cannot form, and the exquisitely spatially-organized CICR that powers every heartbeat becomes weak and asynchronous. Because contractility depends on this microdomain, the ventricle cannot generate adequate force, dilates, and fails.

The genetics reinforce the mechanism: heterozygous LOF is tolerated (gnomAD LOEUF 0.89; unaffected carriers), so **one functional allele suffices** — the disease is unmasked only by biallelic loss, explaining its recessive inheritance and enrichment in consanguineous/founder settings. This is the mirror image of dominant JPH2 **missense** disease (HCM/AF; [PMID: 28393127](#), [PMID: 23973696](#)), where a mutant protein perturbs RyR2 regulation in a dose-dependent, dominant manner. The dichotomy — *dominant missense* → *hypertrophic/arrhythmic* vs *recessive null* → *dilated/failing* — is a clean genotype–mechanism–phenotype correspondence.

The therapeutic corollary follows directly: a loss-of-function disease is the archetypal target for **gene replacement**, and AAV9-JPH2 rescue in mice ([PMID: 27760414](#)) shows that restoring the protein restores dyad structure and Ca^{2+} handling. Until such therapy is available for humans, the clinical pathway is heart-failure GDMT → mechanical support → transplant, for which contemporary pediatric outcomes are strong ([PMID: 42334151](#)).

Evidence Base

PMID	Role	How it supports the findings
30384889	Gene discovery	Identifies JPH2 (homozygous Q428X) in severe childhood cardiomyopathy; establishes CMD2E gene (F001)
31227780	Gene confirmation + population genetics	Homozygous p.E641* recessive DCM; GME carrier/LOF frequencies; founder effect (F001, F004, F006)
33092464	Mechanism	JPH2 joining region binds LTCC to assemble dyad and maintain CICR (F002)
30409805	Mechanism (proteolysis/signaling)	Calpain cleavage disrupts EC-coupling; JP2NT transcription regulator branch (F002)
31506724	Mechanism (proteolysis)	MMP-2 cleaves JPH2, damages dyads in ischemia-reperfusion (F002)
10949023	Model organism	Germline Jph2-null: embryonic lethal, deficient junctional complexes, abnormal Ca ²⁺ (F003)
12086916	Model organism	JP-2 and RyR2 knockouts lethal at heartbeat onset; dyad essential for Ca ²⁺ homeostasis (F003)
21339484	Model organism	Cardiac JPH2 knockdown → contractile failure, heart failure, mortality (F003)
27760414	Therapeutic rationale	AAV9-JPH2 rescues contractility, T-tubules, RyR2 Ca ²⁺ release (F005)
42334151	Clinical management	Excellent 3-yr survival with VAD/transplant in adolescent DCM (F005)
28393127	Contrast (dominant)	JPH2 A405S dominant HCM — distinguishes from recessive CMD2E
23973696	Contrast (dominant)	JPH2 E169K AF via impaired RyR2 stabilization — dominant mechanism
34690801	Human failing heart	RyR2/JPH2 nanoscale disorganization and collagen in failing myocardium

Limitations and Knowledge Gaps

1. **Very small human evidence base.** CMD2E rests on a handful of families (Finnish Q428X; Iranian founder E641*; subsequent case reports). Prevalence, incidence, precise penetrance, sex ratio, and natural-history statistics are therefore not robustly quantified.
 2. **No standardized QoL or registry data** specific to CMD2E; morbidity/QoL is inferred from general pediatric DCM.
 3. **Mechanism largely extrapolated from models and dominant-disease studies.** Direct functional validation of the specific recessive human LOF variants (e.g., patient iPSC-cardiomyocytes) is limited; the causal chain steps 6–7 (cell → organ; arrhythmia/death) are inferred.
 4. **No human gene-therapy data.** AAV9-JPH2 rescue is preclinical (mouse pressure-overload HF), not tested in congenital biallelic-LOF disease or in humans.
 5. **Modifier genes, epigenetics, and environmental interactions** are essentially uncharacterized for CMD2E.
 6. **Ontology mappings** for the specific CMD2E entity (e.g., exact Orphanet subtype, ICD-11 extension code) are approximate; identifiers here reflect the parent DCM class where a subtype-specific code is unavailable.
-

Proposed Follow-up Experiments / Actions

1. **Patient-derived iPSC-cardiomyocyte modeling** of p.Q428X and p.E641* (and isogenic corrected controls) to directly quantify dyad density, LTCC–RyR2 colocalization (dSTORM), Ca²⁺-transient synchrony, and contractility — closing the inference gap between genotype and cellular phenotype.
2. **Conditional / humanized biallelic-LOF mouse** (bypassing embryonic lethality) to establish a postnatal CMD2E model recapitulating the human dilated, progressive course.
3. **Preclinical AAV9-JPH2 gene-replacement trial** in a biallelic-LOF model, with dose-finding and durability endpoints, to advance the rational targeted therapy toward first-in-human studies.
4. **International registry / GeneMatcher-style aggregation** of biallelic *JPH2* DCM cases to define prevalence, penetrance, natural history, and genotype–phenotype correlations.

5. **Targeted carrier screening** for *JPH2* LOF founder alleles in Greater Middle Eastern / consanguineous populations, with genetic counseling and reproductive-option education.
 6. **Cascade testing protocols** and early echocardiographic surveillance guidelines for genotype-positive relatives.
 7. **Deep clinical phenotyping** (arrhythmia burden via Holter/ICD data, cardiac MRI fibrosis) in future cases to refine risk stratification and timing of VAD/transplant.
-

Evidence source types represented: human clinical (case reports/kindreds, pediatric HF registries), model organism (mouse knockout/knockdown/AAV rescue), in vitro/biochemical (JPH2–LTCC interaction, proteolysis), and computational/population genetics (gnomAD constraint, founder-haplotype analysis).

Generated by OpenScientist — Scientific Hypothesis Agent for Novel Discovery